

Lahore Garrison University

DHA Phase 6 Lahore

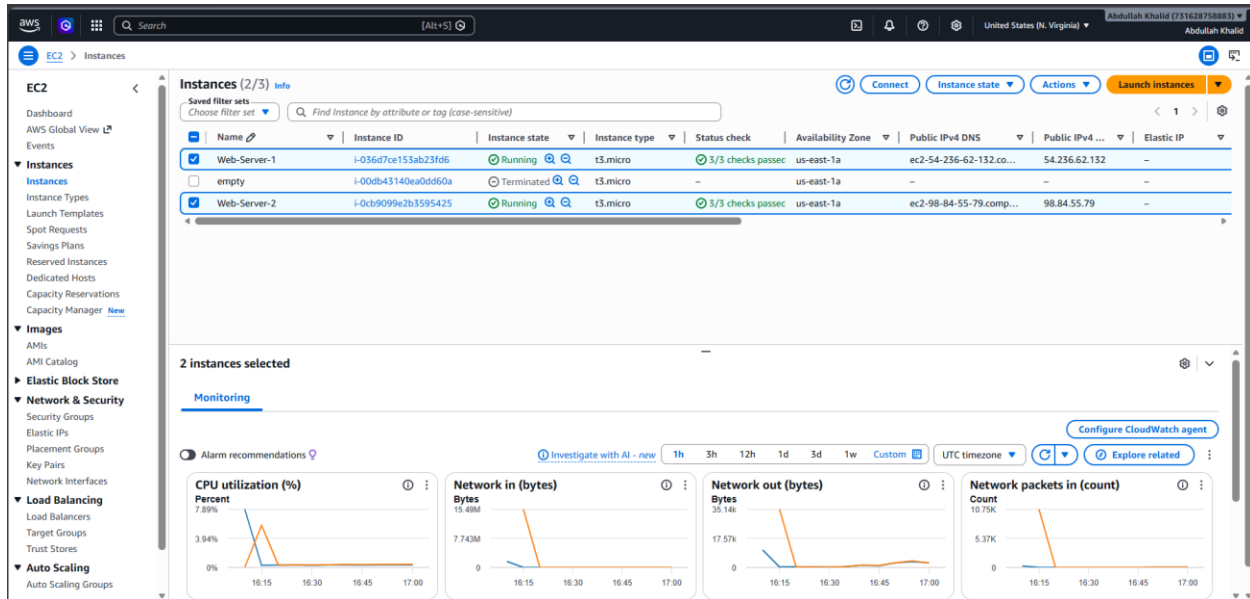


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Subject	Cloud Computing
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Lab 14

Load Balancing using Application Load Balancer

2 Instances created for this lab:



Security groups created:

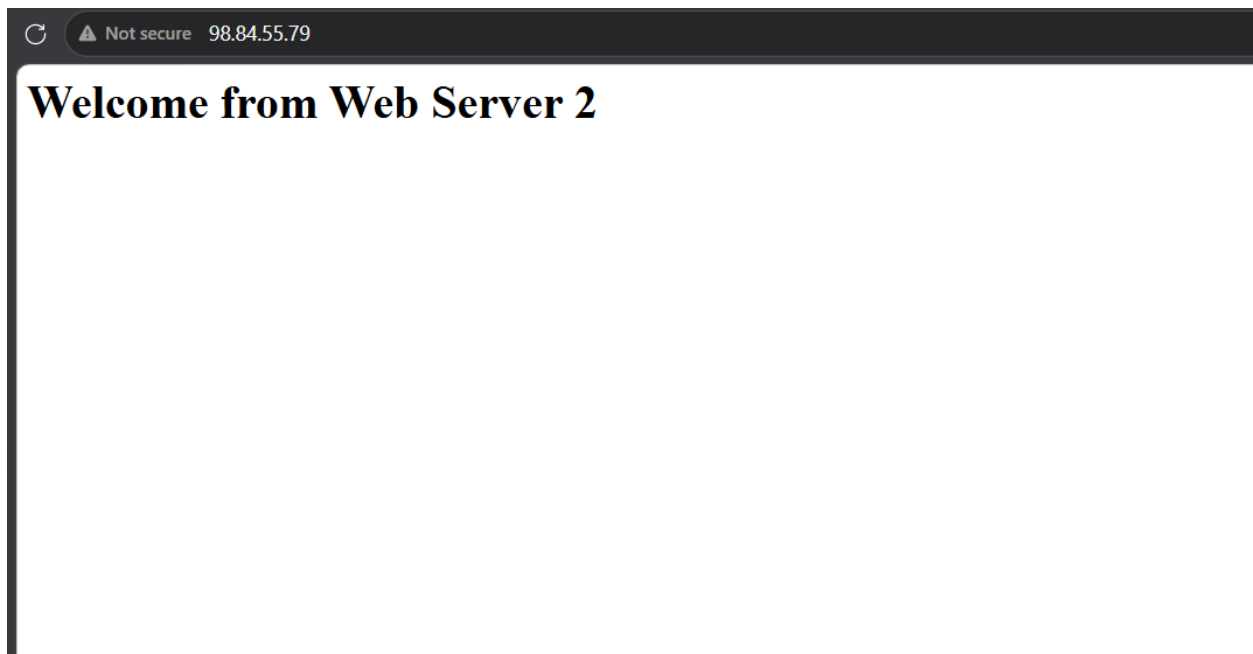
The screenshot displays the AWS Management Console for the 'Security Groups' page. The left sidebar shows the navigation menu with 'Security Groups' selected. The main content area shows a list of 5 security groups. The 'ALB-SG' security group is selected.

Name	Security group ID	Security group name	VPC ID	Description	Owner
launch-wizard-1	sg-0310639bbe41b864f	launch-wizard-1	vpc-03189cced6f1d06d6	launch-wizard-1 created 2026-02-20T0...	731628758883
EC2-Web-SG	sg-034e37aabc01e3194	EC2-Web-SG	vpc-03189cced6f1d06d6	Allows HTTP from ALB and SSH from m...	731628758883
launch-wizard-2	sg-0c8c58b4ff450c01	launch-wizard-2	vpc-03189cced6f1d06d6	launch-wizard-2 created 2026-05-05T1...	731628758883
ALB-SG	sg-0bb511a74a2fb97ae	ALB-SG	vpc-03189cced6f1d06d6	Allows HTTP traffic to Application Load ...	731628758883
default	sg-0a93e1927ef8d42cd	default	vpc-03189cced6f1d06d6	default VPC security group	731628758883

Tested Instance 1 on web server:



Testing Instance 2 on web server:



Created Target Group for both instances:

The screenshot shows the AWS Management Console interface for a Target Group named 'Web-TG'. The left sidebar contains navigation links for various AWS services. The main content area displays the 'Details' tab for the target group, which is associated with the VPC 'vpc-03189cced6f1d06d6'. It shows 2 total targets, all of which are 'Healthy'. Below the details, the 'Targets' tab is active, showing a table of registered targets. Two targets are listed: 'Web-Server-2' and 'Web-Server-1', both with a health status of 'Normal'.

Target type	Protocol : Port	Protocol version	VPC
Instance	HTTP: 80	HTTP1	vpc-03189cced6f1d06d6

2	0	0	2	0	0
Total targets	Healthy	Unhealthy	Unused	Initial	Draining
	0 Anomalous				

Instance ID	Name	Port	Zone	Health	Health st...	Admini...	Overr...	Launch...	Anomaly detection result
i-0cb9099e2b3595425	Web-Server-2	80	us-east-1...	Unused	Target gro...	-	-	May 22, 2...	Normal
i-036d7ce153ab23f66	Web-Server-1	80	us-east-1...	Unused	Target gro...	-	-	May 22, 2...	Normal

Created Application Load Balancer:

The screenshot shows the AWS Management Console interface for an Application Load Balancer named 'Web-ALB'. The left sidebar contains navigation links for various AWS services. The main content area displays the 'Details' tab for the load balancer, which is associated with the VPC 'vpc-03189cced6f1d06d6'. It shows the load balancer is in a 'Provisioning' status and is hosted in the 'us-east-1' region. Below the details, the 'Listeners and rules' tab is active, showing a table of listeners. One listener is listed: 'HTTP:80', which is configured to forward traffic to the target group 'Web-TG'.

Load balancer type	Status	VPC	Load balancer IP address type
Application	Provisioning	vpc-03189cced6f1d06d6	IPv4

Scheme	Hosted zone	Availability Zones	Date created
Internet-facing	Z355XD0TRQ7X7K	subnet-0bb5350350c6298ef us-east-1b (use1-az2) subnet-012488dec6fc88565 us-east-1a (use1-az1)	May 22, 2026, 21:44 (UTC+05:00)

Protocol/Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Tru
HTTP:80	Forward to target group Web-TG: 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not

Observing Target Group instances after creating load balance:

The screenshot shows the AWS Management Console interface for a Target Group named 'Web-TG'. The left sidebar contains navigation links for various AWS services. The main content area displays the 'Details' tab for the target group, which is associated with the VPC 'vpc-03189cced6f1d06d6'. It shows 2 total targets, all of which are 'Healthy'. Below the details, the 'Targets' tab is active, showing a table of registered targets. Two targets are listed: 'Web-Server-2' and 'Web-Server-1', both with a health status of 'Normal'.

Target type	Protocol : Port	Protocol version	VPC
Instance	HTTP: 80	HTTP1	vpc-03189cced6f1d06d6

2	0	0	2	0	0
Total targets	Healthy	Unhealthy	Unused	Initial	Draining
	0 Anomalous				

Observed that both of the instances are healthy:

The screenshot shows the AWS Management Console interface for a Target Group named 'Web-TG'. The left sidebar contains navigation links for EC2, Instances, Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area displays the 'Web-TG' details, including its ARN, target type (Instance), IP address type (IPv4), protocol (HTTP), and port (80). It also shows the VPC ID and a link to the associated Load Balancer. A summary section indicates 2 total targets, all of which are healthy. Below this, a table titled 'Distribution of targets by Availability Zone (AZ)' shows the targets are distributed across two availability zones. The 'Targets' tab is selected, showing a list of registered targets. Both targets are in a 'Healthy' state.

Instance ID	Name	Port	Zone	Health status	Health st...	Admini...	Override details	Launch...	Anomaly detectio...
i-0cb9099e2b3595425	Web-Server-2	80	us-east-1a (us...)	Healthy	-	No override.	No override is curr...	May 22, 2...	Normal
i-036d7ce153ab23fd6	Web-Server-1	80	us-east-1a (us...)	Healthy	-	No override.	No override is curr...	May 22, 2...	Normal

Finally going to test Load balance through browser:

The screenshot shows a web browser window with the address bar displaying 'web-alb-2000357997.us-east-1.elb.amazonaws.com'. The page content displays 'Welcome from Web Server 1'.

After refreshing it shows:

The screenshot shows the same web browser window after a refresh. The address bar remains the same, but the page content now displays 'Welcome from Web Server 2', indicating that the load balancer has successfully routed the request to a different instance.

Lab Question 1: Stop One Instance and Observe Traffic Behavior

I have created target group for my 2 instances and after applying load balancing.

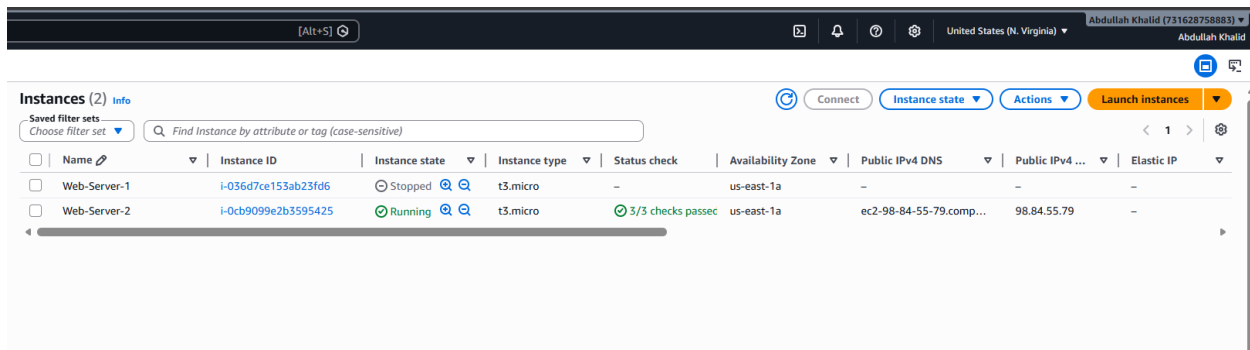
Then tested the load balance on browser using **DNS link** of the load balancer application.

Then attached screenshots of that live working.

By refreshing, it sometimes shows **Welcome from web server 1** and sometimes it shows **Welcome from Web Server 2**.

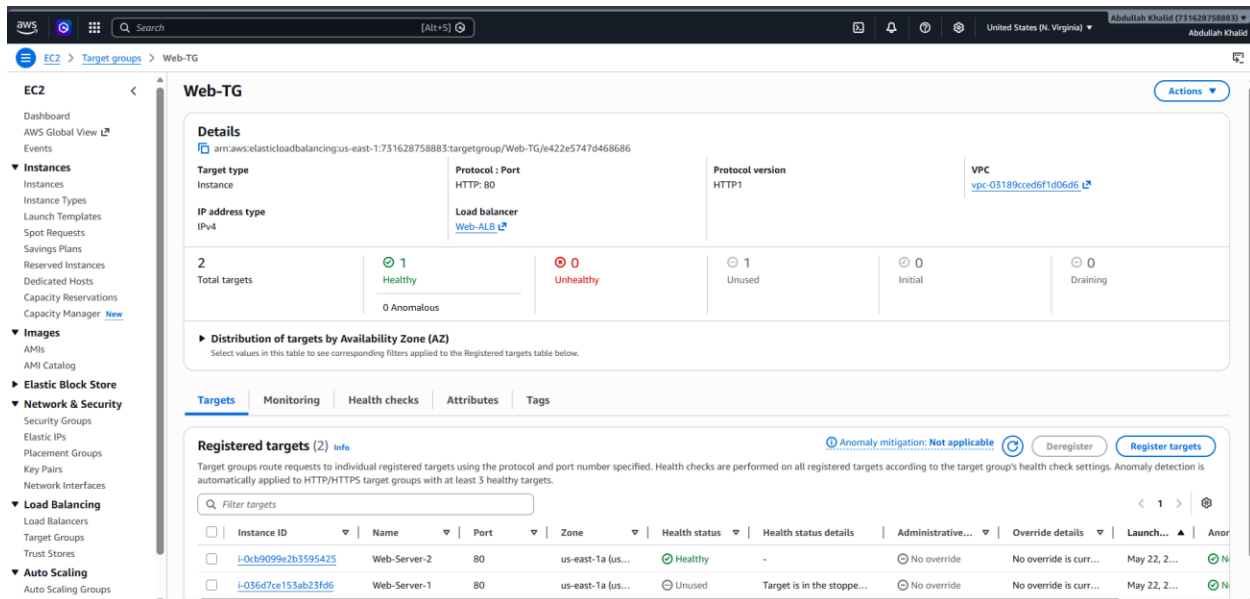
➤ Now I will Stop one Instance and will **observer traffic behavior**.

Instance 1 stopped:



Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
Web-Server-1	i-036d7ce153ab23fd6	Stopped	t3.micro	3/3 checks passed	us-east-1a	-	-	-
Web-Server-2	i-0cb9099e2b3595425	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-98-84-55-79.comp...	98.84.55.79	-

Target group is showing Web-Server-01 as unused:



Instance ID	Name	Port	Zone	Health status	Health status details	Administrative...	Override details	Launch...	Anor
i-0cb9099e2b3595425	Web-Server-2	80	us-east-1a (us...)	Healthy	-	No override	No override is curr...	May 22, 2...	Green
i-036d7ce153ab23fd6	Web-Server-1	80	us-east-1a (us...)	Unused	Target is in the stoppe...	No override	No override is curr...	May 22, 2...	Green

Website is only showing from Web-Server-02 even after refreshing:



Conclusion:

When one EC2 instance was stopped, the Application Load Balancer detected that the instance was unhealthy through health checks. The load balancer stopped sending traffic to the stopped instance and continued forwarding requests to the remaining healthy instance. This demonstrated high availability and fault tolerance.

Lab Question 2: Modify Health Check Path and Test Response

Instance 1 is started again and both are showing healthy now:

The screenshot shows the AWS Management Console for a target group named 'Web-TG'. The 'Details' section displays the following information:

- Target type: Instance
- Protocol : Port: HTTP: 80
- Protocol version: HTTP1
- VPC: vpc-03189cced6f1d06d6
- IP address type: IPv4
- Load balancer: Web-ALB

The 'Targets' section shows 2 total targets, all in a 'Healthy' state. The 'Distribution of targets by Availability Zone (AZ)' section shows 0 Anomalous targets.

Modified Health check path and both instances go on Unhealthy state:

The screenshot shows the AWS Management Console for the 'Web-TG' target group, specifically the 'Health checks' tab. The 'Health check settings' section displays the following information:

- Protocol: HTTP
- Path: /wrong.html (highlighted with a red box)
- Port: Traffic port
- Unhealthy threshold: 2 consecutive health check failures
- Timeout: 5 seconds
- Interval: 30 seconds
- Healthy threshold: 5 consecutive health check successes
- Success codes: 200

Tested the website link and it still works:

The first screenshot shows a web browser displaying 'Welcome from Web Server 1' at the URL 'web-alb-2000357997.us-east-1.elb.amazonaws.com'. The second screenshot shows the same browser displaying 'Welcome from Web Server 2' at the same URL.

Health check path is normal again and instances are healthy:

The screenshot shows the AWS Management Console for a Target Group named 'Web-TG'. The 'Details' section indicates that there are 2 healthy targets and 0 unhealthy targets. The 'Registered targets' table shows two instances, both with a 'Healthy' status.

Instance ID	Name	Port	Zone	Health status	Health status details	Administrative...	Override details	Launch...	Anor
i-036d7ce153ab23fd6	Web-Server-1	80	us-east-1a (us...)	Healthy	-	No override	No override is curr...	May 22, 2...	Healthy
i-0cb9099e2b3595425	Web-Server-2	80	us-east-1a (us...)	Healthy	-	No override	No override is curr...	May 22, 2...	Healthy

Testing on web server:

The first screenshot shows a web browser displaying 'Welcome from Web Server 1'. The second screenshot shows a web browser displaying 'Welcome from Web Server 2'.

Conclusion:

The health check path was changed from /health.html to /wrong.html. Since /wrong.html did not exist on the EC2 instances, the load balancer health checks failed and the targets became unhealthy. After changing the path back to /health.html, the targets passed health checks and became healthy again. This shows that health checks are important for routing traffic only to working application endpoints.

Lab Question 3: Attach Auto Scaling Group and Monitor Scaling

Created Launch Template:

The screenshot shows the AWS Management Console interface for a Launch Template named 'Web-LT' (ID: lt-02ac6e5fd34dc8940). The console is in the 'United States (N. Virginia)' region, and the user is 'Abdullah Khalid'.

Launch template details

- Launch template ID: lt-02ac6e5fd34dc8940
- Launch template name: Web-LT
- Default version: 2
- Owner: arn:aws:iam::731628758883:root

Launch template version details

Version: 2 (Default)

Description: -

Date created: 2026-05-22T17:52:09.000Z

Created by: arn:aws:iam::731628758883:root

Instance details

- AMI ID: ami-0ed094fb1304fd857
- Instance type: t3.micro
- Availability Zone: -
- Availability Zone Id: -
- Key pair name: lab1-key
- Security groups: -
- Security group IDs: sg-054e57aabc01e3194

Auto Scaling group created:

The screenshot shows the AWS Management Console interface for an Auto Scaling group named 'Web-ASG'. The console is in the 'United States (N. Virginia)' region, and the user is 'Abdullah Khalid'.

Auto Scaling groups (1)

Search your Auto Scaling groups

Launch configurations | Launch templates | Actions | Create Auto Scaling group

Web-ASG | At desired capacity | Web-LT | Version Default | 2 | 2/2 Healthy | 2 | 2 | 4 | 2 Availability Zones | Fri May 22 2020

0 Auto Scaling groups selected

Select an Auto Scaling group

ASG Activity showing instance launch:

Activity history (2)				
Filter activity history Filter by a date and time range				
Status	Description	Cause	Start time	End time
Successful	Launching a new EC2 instance: i-096ce15ce164195df	At 2026-05-22T17:58:21Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-05-22T17:58:23Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 May 22, 10:58:25 PM +05:00	2026 May 22, 10:58:30 PM +05:00
Successful	Launching a new EC2 instance: i-0ceefbd735134f7d9	At 2026-05-22T17:58:21Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-05-22T17:58:23Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 May 22, 10:58:25 PM +05:00	2026 May 22, 10:58:32 PM +05:00

EC2 Instances showing ASG-created instances:

Instances (5)									
Save filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)									
☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐		i-096ce15ce164195df	Running	t3.micro	3/3 checks passed	us-east-1b	ec2-100-24-33-109.co...	100.24.33.109	-
☐		i-07d922102c70af526	Running	t3.micro	3/3 checks passed	us-east-1b	ec2-54-204-206-25.co...	54.204.206.25	-
☐	Web-Server-1	i-036d7ce153ab23fd6	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-98-92-217-175.co...	98.92.217.175	-
☐		i-0ceefbd735134f7d9	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-100-54-122-85.co...	100.54.122.85	-
☐	Web-Server-2	i-0cb9099e2b3595425	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-98-84-55-79.comp...	98.84.55.79	-

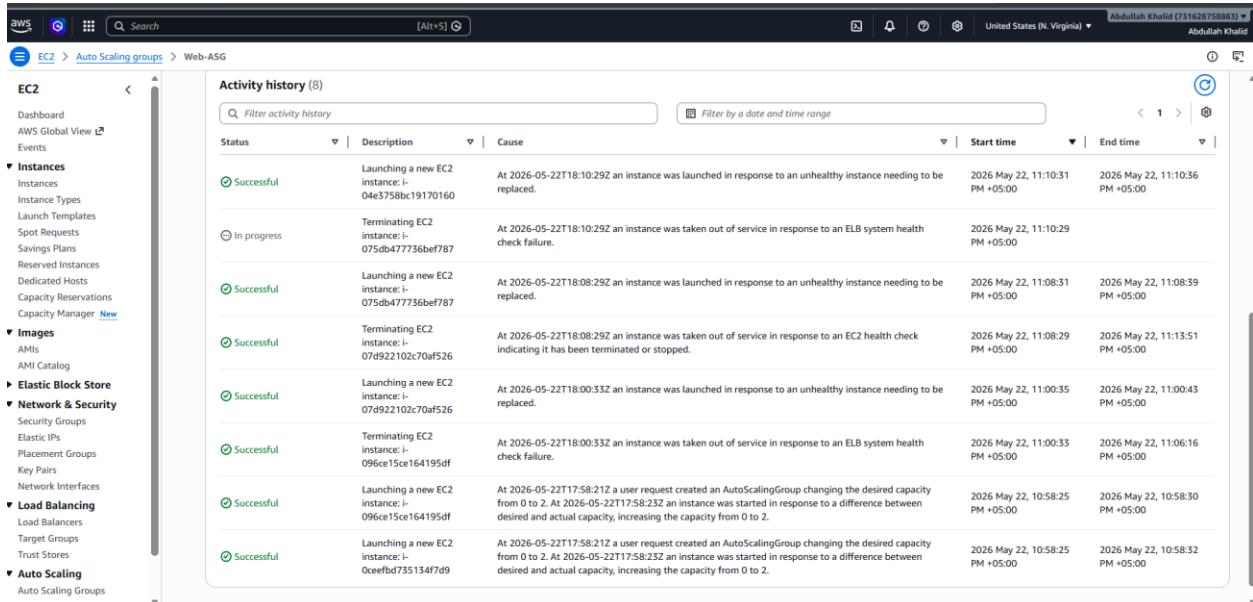
Target group showing ASG instances healthy:

Web-TG									
Details arn:aws:elasticloadbalancing:us-east-1:731628758883:targetgroup/Web-TG/e422e5747d468686 Target type: Instance Protocol: Port: HTTP: 80 Protocol version: HTTP1 VPC: vpc-03189ced6f1d06d6 IP address type: IPv4 Load balancer: Web-ALB 4 Total targets 4 Healthy 0 Unhealthy 0 Unused 0 Initial 0 Draining 0 Anomalous Distribution of targets by Availability Zone (AZ) Select values in this table to see corresponding filters applied to the Registered targets table below.									

Terminated one instance created by ASG to observe next behavior:

Instances (1/7)									
Save filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)									
☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐	Web-Server-1	i-036d7ce153ab23fd6	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-98-92-217-175.co...	98.92.217.175	-
☐		i-0ceefbd735134f7d9	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-100-54-122-85.co...	100.54.122.85	-
☐	Web-Server-2	i-0cb9099e2b3595425	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-98-84-55-79.comp...	98.84.55.79	-
☒		i-096ce15ce164195df	Terminated	t3.micro	-	us-east-1b	-	-	-
☐		i-075db47736bef787	Running	t3.micro	3/3 checks passed	us-east-1b	ec2-54-157-226-236.co...	54.157.226.236	-
☐		i-07d922102c70af526	Terminated	t3.micro	-	us-east-1b	-	-	-
☐		i-04e3758bc19170160	Running	t3.micro	3/3 checks passed	us-east-1b	ec2-13-218-19-35.com...	13.218.19.35	-

More Instances created using ASG after terminating from them:



Status	Description	Cause	Start time	End time
Successful	Launching a new EC2 instance: i-04e3758bc19170160	At 2026-05-22T18:10:29Z an instance was launched in response to an unhealthy instance needing to be replaced.	2026 May 22, 11:10:31 PM +05:00	2026 May 22, 11:10:36 PM +05:00
In progress	Terminating EC2 instance: i-075db477736bef787	At 2026-05-22T18:10:29Z an instance was taken out of service in response to an ELB system health check failure.	2026 May 22, 11:10:29 PM +05:00	
Successful	Launching a new EC2 instance: i-075db477736bef787	At 2026-05-22T18:08:29Z an instance was launched in response to an unhealthy instance needing to be replaced.	2026 May 22, 11:08:31 PM +05:00	2026 May 22, 11:08:39 PM +05:00
Successful	Terminating EC2 instance: i-07d922102c70af526	At 2026-05-22T18:08:29Z an instance was taken out of service in response to an EC2 health check indicating it has been terminated or stopped.	2026 May 22, 11:08:29 PM +05:00	2026 May 22, 11:13:51 PM +05:00
Successful	Launching a new EC2 instance: i-07d922102c70af526	At 2026-05-22T18:00:33Z an instance was launched in response to an unhealthy instance needing to be replaced.	2026 May 22, 11:00:35 PM +05:00	2026 May 22, 11:00:43 PM +05:00
Successful	Terminating EC2 instance: i-096ce15ce164195df	At 2026-05-22T18:00:33Z an instance was taken out of service in response to an ELB system health check failure.	2026 May 22, 11:00:33 PM +05:00	2026 May 22, 11:06:16 PM +05:00
Successful	Launching a new EC2 instance: i-096ce15ce164195df	At 2026-05-22T17:58:21Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-05-22T17:58:23Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 May 22, 10:58:25 PM +05:00	2026 May 22, 10:58:30 PM +05:00
Successful	Launching a new EC2 instance: i-0ceefbd735134f7d9	At 2026-05-22T17:58:21Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-05-22T17:58:23Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 May 22, 10:58:25 PM +05:00	2026 May 22, 10:58:32 PM +05:00

Conclusion:

An Auto Scaling Group was created using a launch template and attached to the Application Load Balancer target group. The ASG maintained the desired number of EC2 instances. When an instance was terminated, the Auto Scaling Group automatically launched a replacement instance and registered it with the target group. This demonstrated automatic recovery, scalability, and high availability.